FAMILY HABIT APP · DATASTORE · DRAFT 3 FOR YOUR REVIEW · 1 NOV 2026

Datastore tables, draft 3

There are **14 new tables** to build now, in **four colour-coded groups**. Six more are designed later. The colours stay the same all the way down this page.

About each parent

`parents` `children` `consents` `picks` `slots`

What happened

`habit_logs` `messages_sent` `connection_answers` `events`

App look up settings

`settings_versions` `event_types` `holiday_dates`

Housekeeping

`invites` `deletion_log`

What changed since draft 2

- Your side-chat review applied, with my three changes you agreed. Highlights: `logs` is renamed `habit_logs`; event types are a list; duplicates from Telegram resends are blocked; tables point at the nudge's ID number.
- New tables: `children` (birth year, optional), `connection_answers` (metric 10), `event_types`, `holiday_dates`, `invites`.
- The four experiment tables move to "before alpha", designed with your start and stop flow. So the third group is renamed from "Experiment control room" to **App look up settings**.
- Opting out of experiments moves from `assignments` to `consents`.
- A photo that fails the check is still stored and shown to the parent; the log is recorded as self-reported.
- The server only adds rows, except three narrow edits, each of which also writes an event.

1. The tables, one by one
2. Which values can I change, and how to change them?
3. How to visualise if 2 or more experiments are running at the same time?
4. Decisions: made, still open, parked

PART 1

The tables, one by one

Each card shows what **one row** means, its columns in plain words, the **rules the database checks**, and **what our server may do**. Unless a card says otherwise, the server may only add rows. Removing a parent's rows is done only by "delete my data" (rule 1.5).

About each parent

Personal. "Delete my data" removes all of it automatically.

parents

Each row: one parent.

`id` a number the database makes up; not their name

`telegram_id` how the bot finds them (unique)

`first_name` for friendly messages

`signed_up_on` starts the 5-week count (10.3, 11.11)

`cohort` neutral label, e.g. cohort-1 (11.11)

`paused` yes / no (paused = muted)

`wrap_up_on` yes / no, default yes (3.12)

`wrap_up_minute` default 1200 = 8:00pm (3.12)

`created_at` audit time

RULES THE DATABASE CHECKS

- Telegram ID used once only

`SERVER MAY` Add a parent. Edit `ONLY` paused, wrap_up_on and wrap_up_minute, and every edit also writes an event.

children new

Each row: one child of a parent. Optional.

`id`

`parent_id` which parent

`birth_year` e.g. 2017; age worked out within 1 year

`replaces_id` if the parent corrects an answer, the row it replaces

`created_at` when the parent told us

RULES THE DATABASE CHECKS

- Birth year from 25 years ago to this year; anything else is refused at sign-up `AND` in the saving code
- Ages 0 to 6 and 13 to 25 are accepted and flagged in Telemetry (not refused)
- No name, no birth month, no school
- Created now, empty in week one; asked from alpha

`SERVER MAY` Add only.

consents

Each row: one answer a parent gave about permission.

`parent_id`

`topic` recording (11.9), photos (5.5), experiments (opt-out, 11.9)

`answer` agreed / opted out

`wording_version` which sentence they saw

`created_at` when

RULES THE DATABASE CHECKS

- The latest answer per topic counts
- Opt-out lives here now, not in assignments (review point 6)

`SERVER MAY` Add only.

picks

Each row: one behaviour a parent chose.

`parent_id`

`nudge_id` the nudge's ID number (v1.6)

`frequency` e.g. every_day, twice_a_week; checked in code, not a database list

`chosen_at`

`stopped_at` empty while still chosen

RULES THE DATABASE CHECKS

- Only one current pick per parent per nudge
- stopped_at can't be before chosen_at
- A nudge with picks can't be deleted (v1.5 promise)

`SERVER MAY` Add. Fill in stopped_at `ONCE`; that also writes an event.

slots

Each row: one nudge time.

`parent_id`

`schedule` school day / non-school day

`start_minute` minutes after midnight, e.g. 1140 = 7:00pm

`created_at`

RULES THE DATABASE CHECKS

- No duplicate time on the same schedule

`SERVER MAY` Replace a parent's slots when they change them; that also writes an event.

What happened

Personal. Add-only diaries.

habit_logs (renamed)

Each row: a parent says "I did it". (Was `logs`.)

`parent_id, nudge_id`

`for_day` today or yesterday only (5.4), checked by the saving function

`logged_at` when they told us (new)

`from_message_id` the nudge message the Done tap came from; empty = logged on their own

`description` up to 280 characters

`photo_path` link to the photo store, if any

`photo_check` passed / failed / no photo

`recorded_as` photo-verified / self-reported

`telegram_update_id` Telegram's message ID, used once only

RULES THE DATABASE CHECKS

- A photo that `FAILED` the check is still stored and shown to the parent in the Web app; the log is recorded as self-reported (rule 1.3)
- "Counts for which week" is not stored: `week.ts` works it out (one copy of rule 7.6)

SERVER MAY Add only.

messages_sent

Each row: one message the bot started.

`parent_id`

`kind` nudge message / Weekly Wrap Up

`nudge_id, example_id` which nudge and idea (3.13); required for a nudge message

`for_day, slot_minute` the Singapore day and slot

`sent_at`

RULES THE DATABASE CHECKS

- Only one nudge message per parent, day and slot: a backstop if two scheduler runs overlap
- Only one Wrap Up per parent per week

SERVER MAY Add only.

connection_answers (new)

Each row: a parent's weekly 1 to 5 answer to "How connected did you feel with your child?" (metric 10).

`parent_id`

`week_starting` the Monday of that week

`score` 1 to 5

`telegram_update_id` used once only

`created_at`

RULES THE DATABASE CHECKS

- One answer per parent per week (suggested: a second tap is ignored)

SERVER MAY Add only.

events

Each row: anything that happened, for Telemetry.

`parent_id`

`event_type_id` points at `event_types`, not free text

`occurred_at` when

`settings_version` e.g. 3; never empty (11.2)

`variants` structured list, e.g. style-test-1 = playful; "baseline" in week one; never empty (11.2)

`telegram_update_id` used once only, when the event came from Telegram

RULES THE DATABASE CHECKS

- Written in the same step as the log or message it describes (all or nothing)

SERVER MAY Add only.

App look up settings

About the whole app, not any one parent: lists the app looks things up in (a calendar of holidays, a glossary of event types) and the history of everyday settings. Nothing personal; kept when a parent deletes their data.

settings_versions

Each row: one edition of everyone's everyday values.

`version` 1, 2, 3…
`in_force_from`
`note` plain English, from the pull request
`settings` labelled list of every value (11.1)
`fingerprint` a code made from the values, unique

RULES THE DATABASE CHECKS

- Made automatically when new code goes live (question B)
- The fingerprint stops two app copies both creating "version 3"

SERVER MAY Add only. Never edited or deleted.

event_types new

Each row: one kind of event, e.g. "nudge sent", "Not today", "too many nudges".

`id` a small number
`name`
`meaning` plain English
`metric` which of the 10 metrics it feeds

RULES THE DATABASE CHECKS

- A new reward feature adds a row, not a rewrite

SERVER MAY Read only. You add rows through a reviewed script.

holiday_dates new

Each row: one school or public holiday.

`day` a Singapore date, unique
`kind` MOE school holiday / public holiday
`name` e.g. Deepavali

RULES THE DATABASE CHECKS

- Decides the non-school-day schedule (3.2)

SERVER MAY Read only. You upload the dates before launch.

Housekeeping

Nothing personal once a parent is deleted.

invites new

Each row: one invitation (rule 1.1, invite-only).

`code` unique
`created_at`
`used_by` the parent who used it; cleared if they delete their data
`used_at`

RULES THE DATABASE CHECKS

- Each code works once

SERVER MAY You create codes; the server marks one used.

deletion_log

Each row: "Delete my data" happened.

`on_day` e.g. 2 Nov
`how_many` e.g. 1 parent, no name or ID

RULES THE DATABASE CHECKS

- With about 12 parents, a date can hint at who it was. Already agreed; noted only

SERVER MAY Only the "delete my data" function writes here.

Designed later (6 tables)

Not built now. Shown so you can see the whole picture.

`experiments`

one row per experiment, name unique

`Before alpha, designed with your start and stop flow`

`experiment_options`

one row per option and its share, e.g. playful 50%

`Before alpha`

`experiment_changes`

one row per start, change or stop

`Before alpha`

`assignments`

which option each parent got; one per parent per experiment

`Before alpha`

`album_entries`

a saved moment from the Wrap Up (7.7)

`With Variable rewards`

`badges_earned`

one badge a parent earned

`With Variable rewards`

Already live, unchanged: `nudges` and `examples` (the Nudge repository). **Outside the database:** photos in Supabase Storage, and short-term chat memory in Upstash Redis. "Delete my data" reaches both.

Rules for every table

Locked by default

Every table starts with no access. The server is then given only the permissions on its card (rule 8.2), and a test checks them.

All or nothing

Each action, such as "save a log and its event", runs as one database step. It never stops halfway, like a bank transfer.

No duplicates from Telegram

Telegram resends messages it thinks failed. Its message ID can be saved only once.

Delete follows the parent

Every personal table is linked to `parents`, so deleting a parent deletes their rows. A test fails if a new table isn't linked. Order: photos, then chat memory, then rows, then the log.

Structure only in the database

The database checks links, duplicates and the 280-character limit. Tunable numbers such as "3 a day" stay in code (rule 1.4).

Telemetry sees less

Telemetry reads through views with no names, Telegram IDs or descriptions.

Labelled and timed

Every table has a `created_at` time and a plain-English note on every column, stored in the database itself.

Fast lookups

Indexes for the busiest questions: "nudges sent today", "last log per habit", "who has a slot now", and "events by type and date".

Times stored one way

"When" = a time with time zone. Singapore day = a date. Slot = minutes after midnight.

PART 2 · YOUR QUESTION 1

Which values can I change, and how to change them?

2.1 The five kinds of value

Every value in the app is one of five kinds. The kind decides what an experiment is allowed to do with it.

`Tunable`
A number that can be tested higher or lower.

`Named style`
A choice between reviewed wordings or tones.

`Named rule`
A choice between reviewed behaviours.

`Guard-rail`
A safety limit that an experiment may only make stricter.

`Fixed`
Never changes, not even in an experiment (rule 11.4).

2.2 All 29 values

The last column says where each value is in the code today. "Not built yet" means there is nothing to change until that part of the app is built. Checked against the code on 1 Nov.

#	KIND	WHAT IT MEANS	VALUES IT CAN TAKE	IN THE CODE TODAY
`Tunable`				
1	`Tunable`	**Default slot times:** the nudge times a new parent starts with. Parents can always change their own.	Any 30-minute start from 7:00am to 9:30pm. **Today:** 7:00am, 7:00pm and 9:00pm on school days; 9:00am, 12:30pm and 7:00pm on other days	`settings.ts`
2	`Tunable`	**Behaviours a parent may pick:** how many habits a parent works on at once.	Numbers for quick, deliberate and the minimum in total. **Today:** up to 2 quick, up to 1 deliberate, at least 1	`settings.ts`
3	`Tunable`	**Weekly Wrap Up default time:** when the Sunday summary is sent unless the parent changes it.	Any 30-minute start from 7:00am to 9:30pm. **Today:** 8:00pm	`settings.ts`
4	`Tunable`	**Slots per schedule:** how many nudge times a parent can set for each kind of day.	A number, at least 1. **Today:** up to 3	`settings.ts`
5	`Tunable`	**Slot length and range:** how long each nudge window lasts, and the earliest and latest start times.	Minutes and times. **Today:** 30 minutes, 7:00am to 9:30pm	`settings.ts`
6	`Tunable`	**Times a week for deliberate behaviours:** how many reminders a week for the bigger family activities.	A number. **Today:** 2	`settings.ts`
7	`Tunable`	**Examples in a message:** how many ideas a nudge includes.	0, 1 or 2. Added when the message words are built (rule 3.13 says 1 for now)	Not built yet checked: not in the code

#	KIND	WHAT IT MEANS	VALUES IT CAN TAKE	IN THE CODE TODAY
8	Tunable	**Frequencies offered:** the choices parents see for how often to do a habit.	Sets of choices. **Today:** every day or every 2 days (quick); 1 or 2 times a week (deliberate)	`choices.ts` outside settings.ts (rule 1.4, question L)
9	Tunable	**Hidden ideal and floor:** the behind-the-scenes target, and the minimum before the app nudges that habit first.	Numbers. **Floor today:** a quick habit not logged today or yesterday is nudged first; deliberate habits have no floor. **Ideal:** not used in the code yet (written only in EXPERIMENTS.md)	`planner.ts (floor)` outside settings.ts (rule 1.4, question L)
10	Tunable	**Reward numbers:** points per log, the bonus for logging without a nudge, and how many points each level needs.	Numbers. Added when Variable rewards is built (the 2x bonus is in the spec, not the code)	Not built yet
Named style				
11	Named style	**Message style:** the tone of nudge messages.	Named styles, for example warm or playful. Added when the message words are built	Not built yet
12	Named style	**Framing:** whether messages ask the parent to plan ahead or act now.	plan ahead / do now. Added when the message words are built	Not built yet
13	Named style	**Message length and emoji:** how long messages are and whether they use emoji.	Named styles. Added when the message words are built	Not built yet
14	Named style	**Button wording:** the words on the reply buttons.	Named sets. Added when the Telegram bot is built ("Done" and "Not today" are agreed in rule 3.5)	Not built yet
15	Named style	**Thank-you style:** how the bot thanks the parent after a log.	Named styles. Added when Variable rewards is built	Not built yet
Named rule				
16	Named rule	**Days deliberate behaviours are nudged:** which days the bot reminds about bigger activities.	Named patterns. **Today:** Friday and Sunday	`settings.ts`
17	Named rule	**Rest after a streak:** after a quick habit is done several days in a row, the bot nudges about it less often.	Named rules. **Today:** after 7 days in a row, not nudged two days running	`settings.ts (7)` + `planner.ts`
18	Named rule	**Defaults pre-selected or not:** whether new parents start with suggested times or pick their own.	pre-selected / pick from scratch. Added when the setup screen is built (agreed: pre-selected)	Not built yet

#	KIND	WHAT IT MEANS	VALUES IT CAN TAKE	IN THE CODE TODAY
19	Named rule	**Priority order:** which habit gets nudged first when slots are limited.	Named orders. **Today:** quick habits below the floor first (the one not logged for longest goes first), then other quick, then deliberate; a tie goes to the habit picked first	`planner.ts`
20	Named rule	**Timing within a slot:** when, inside the 30-minute window, the nudge arrives.	at the start / a random minute. Added when the message sender is built	Not built yet checked: not in the code
21	Named rule	**Use of the parent's own context:** whether messages mention the parent's past notes or their child's interests.	on / off. Added when the message words are built	Not built yet
22	Named rule	**Photo or description prompt:** what the bot asks for after a log.	photo / description / both / neither. Added when the Telegram bot is built	Not built yet
23	Named rule	**Claude model used:** which AI model writes the messages, a trade-off between cost and quality.	Named models. Added when the message words are built	Not built yet
24	Named rule	**Badges and streak rules:** how badges are earned and how many missed days a streak survives.	Named rules. Added when Variable rewards is built (spec: a streak survives 2 missed days)	Not built yet
Guard-rail				
25	Guard-rail	**Nudge messages per day:** the most nudge messages a parent can get in one day. Replies and the Weekly Wrap Up don't count.	A number, 3 or fewer (stricter = fewer). **Today:** 3	`settings.ts`
26	Guard-rail	**Gap between slots:** the shortest time allowed between two nudge times on the same kind of day.	1 hour or longer (stricter = longer). **Today:** 1 hour	`settings.ts`
Fixed				
27	Fixed	**Note length limit:** the longest description a parent can write when logging.	Always 280 characters	`database (nudges table)`
28	Fixed	**Time zone:** every day, week and time in the app follows Singapore time.	Always Singapore time, midnight to midnight, Monday to Sunday weeks	`sgt.ts`
29	Fixed	**At least 1 slot:** every parent must have at least one nudge time.	Always at least 1	`schedule.ts`

> **Found while checking:** rule 1.4 says every adjustable number lives once, in `src/settings.ts`. Two don't yet: the frequency choices (row 8, in `choices.ts`) and the 2-day floor (row 9, in `planner.ts`). Both work; they are in the wrong file. See question **L**.

2.3 Where experiment information is kept

Read left to right. The code holds the values. The database holds which option is in force, who got it, and what each parent actually had at every moment.

NOT A TABLE: THE CODE

The list of values

Every value in 2.2, its everyday setting, and the other options you have approved to try.

Changed by: a pull request you approve

↓

SETTINGS, NOW · EXPERIMENTS, BEFORE ALPHA

`settings_versions`

A saved copy of everyone's everyday values, numbered 1, 2, 3 each time they change.

Written by: the app, automatically · built now

`experiments`

+

`experiment_options`

+

`experiment_changes`

Each experiment once; its options and split; and each start, change or stop.

Written by: you · before alpha

`assignments`

Which option each parent got. It never changes.

Written by: the app, at random · before alpha

↓

WHAT HAPPENED

`events`

Every event is stamped with the settings version and the parent's options. In week one the stamp says `baseline`.

Written by: the app, every time · built now

2.4 One experiment, followed through

A story in six steps: you want to test whether **playful** nudge messages work better than **warm** ones. Each step shows the table that changes, with example rows. The coloured edge tells you which table group it belongs to.

NAMES USED IN EVERY STEP

Experiment: style-test-1 **Value changed:** Message style (row 11) **Control (Option A): warm** **Test (Option B): playful**

Settings version: version 1 **Parents:** P-07 (Test) · P-08 (Control) · P-09 (opted out)

1 You approve the Test option

You · pull request · in the code, not a table

The code lists all 29 values. For each one: the **Control (Option A)**, which is the everyday value every parent normally gets, and the **Test (Option B)**, an alternative you have approved for an experiment. This experiment changes only **row 11, message style** (highlighted). Every other value is not tested, so all parents get its Control value.

#	Value	Kind	Control (Option A)	Test (Option B)
1	Default slot times	Tunable	School days 7:00am, 7:00pm, 9:00pm; other days 9:00am, 12:30pm, 7:00pm	Not tested, same as Control
2	Behaviours a parent may pick	Tunable	2 quick, 1 deliberate, at least 1	Not tested, same as Control
3	Weekly Wrap Up default time	Tunable	8:00pm	Not tested, same as Control
4	Slots per schedule	Tunable	up to 3	Not tested, same as Control
5	Slot length and range	Tunable	30 minutes, 7:00am to 9:30pm	Not tested, same as Control
6	Times a week for deliberate behaviours	Tunable	2	Not tested, same as Control
7	Examples in a message	Tunable	not built yet	Not tested, same as Control
8	Frequencies offered	Tunable	every day / every 2 days; 1 or 2 a week	Not tested, same as Control
9	Hidden ideal and floor	Tunable	floor: not logged today or yesterday	Not tested, same as Control
10	Reward numbers	Tunable	not built yet	Not tested, same as Control
11	Message style ← **The value being changed in this experiment.** Control parents get **warm** messages; Test parents get **playful** ones.	Named style	**warm** (planned; built with the message words)	**playful**

#	Value	Kind	Control (Option A)	Test (Option B)
12	Framing	Named style	not built yet	Not tested, same as Control
13	Message length and emoji	Named style	not built yet	Not tested, same as Control
14	Button wording	Named style	not built yet	Not tested, same as Control
15	Thank-you style	Named style	not built yet	Not tested, same as Control
16	Days deliberate behaviours are nudged	Named rule	Friday and Sunday	Not tested, same as Control
17	Rest after a streak	Named rule	after 7 days in a row, not nudged two days running	Not tested, same as Control
18	Defaults pre-selected or not	Named rule	not built yet	Not tested, same as Control
19	Priority order	Named rule	below the floor, then other quick, then deliberate	Not tested, same as Control
20	Timing within a slot	Named rule	not built yet	Not tested, same as Control
21	Use of the parent's own context	Named rule	not built yet	Not tested, same as Control
22	Photo or description prompt	Named rule	not built yet	Not tested, same as Control
23	Claude model used	Named rule	not built yet	Not tested, same as Control
24	Badges and streak rules	Named rule	not built yet	Not tested, same as Control
25	Nudge messages per day	Guard-rail	3	Not tested, same as Control

#	Value	Kind	Control (Option A)	Test (Option B)
26	Gap between slots	`Guard-rail`	1 hour	Not tested, same as Control
27	Note length limit	`Fixed`	280 characters	can never be tested
28	Time zone	`Fixed`	Singapore time	can never be tested
29	At least 1 slot	`Fixed`	at least 1	can never be tested

- **"Not tested, same as Control"** means this value isn't part of the experiment, so Test parents get exactly the same value as Control parents.
- **"Can never be tested"** marks the three Fixed values (rule 11.4).
- Control shows today's real values from the code; "not built yet" means that part of the app doesn't exist yet. In the code, the Test column is called `options_approved_to_try`, because you could approve more than one Test option (B, C…).

2 Everyone starts on the same everyday values

App, automatically · `settings_versions` · App look up settings

The app keeps a saved copy of the **Control (Option A)** column from step 1 (everyone's everyday values), called a **settings version**. Right now that is **version 1**. **Both groups in this experiment are on version 1.** The experiment doesn't change the version; it sits on top of it, swapping just one value (row 11) for the parents who get the Test (Option B).

Version 1

Everyone's everyday values (Control)

`29 values`

1	Default slot times	
2	Behaviours a parent may pick	
3	Weekly Wrap Up default time	
4	Slots per schedule	
5	Slot length and range	
6	Times a week for deliberate behaviours	
7	Examples in a message	
8	Frequencies offered	
9	Hidden ideal and floor	
10	Reward numbers	
11	Message style	`warm`
12	Framing	
13	Message length and emoji	
14	Button wording	
15	Thank-you style	
16	Days deliberate behaviours are nudged	
17	Rest after a streak	
18	Defaults pre-selected or not	
19	Priority order	
20	Timing within a slot	
21	Use of the parent's own context	
22	Photo or description prompt	
23	Claude model used	
24	Badges and streak rules	
25	Nudge messages per day	
26	Gap between slots	
27	Note length limit	
28	Time zone	
29	At least 1 slot	

Row 11 is the value this experiment changes. Test parents get playful instead; the version itself does not change.

▼ See all 29 values in version 1

#	Value	Version 1
1	Default slot times	School days 7:00am, 7:00pm, 9:00pm; other days 9:00am, 12:30pm, 7:00pm
2	Behaviours a parent may pick	2 quick, 1 deliberate, at least 1
3	Weekly Wrap Up default time	8:00pm
4	Slots per schedule	up to 3
5	Slot length and range	30 minutes, 7:00am to 9:30pm

#	Value	Version 1
6	Times a week for deliberate behaviours	2
7	Examples in a message	not built yet
8	Frequencies offered	every day / every 2 days; 1 or 2 a week
9	Hidden ideal and floor	floor: not logged today or yesterday
10	Reward numbers	not built yet
11	Message style	warm (planned)
12	Framing	not built yet
13	Message length and emoji	not built yet
14	Button wording	not built yet
15	Thank-you style	not built yet
16	Days deliberate behaviours are nudged	Friday and Sunday
17	Rest after a streak	after 7 days in a row, not nudged two days running
18	Defaults pre-selected or not	not built yet
19	Priority order	below the floor, then other quick, then deliberate
20	Timing within a slot	not built yet
21	Use of the parent's own context	not built yet
22	Photo or description prompt	not built yet
23	Claude model used	not built yet
24	Badges and streak rules	not built yet
25	Nudge messages per day	3
26	Gap between slots	1 hour
27	Note length limit	280 characters
28	Time zone	Singapore time
29	At least 1 slot	at least 1

A new version is made only when you change an everyday value **for everyone**. That is shown in 2.5, way A, not in this story.

3 You start the experiment

`You · three experiment tables · before alpha`

Three small tables, each with one job: **which value** is tested, the **Control and Test groups** and how parents are split between them, and **when** it started and stopped.

`experiments` : which value is tested

Experiment	Value changed
style-test-1	Message style (row 11)

`experiment_options` : Control and Test, and the split

Experiment	Group	Message style	Share of parents
style-test-1	Control (Option A)	warm	50%
style-test-1	Test (Option B)	playful	50%

`experiment_changes` : when

Experiment	Action	When	Note
style-test-1	start	Week 1	
style-test-1	stop	Week 7	Enough data

4 The app puts each parent in Control or Test, at random

`App · assignments · before alpha`

This is **at the parent level**: which group each parent is in.

- One row per parent per experiment. The group **never changes** while the experiment runs (rule 11.6); the only exceptions are opting out and "delete my data".
- A parent in two experiments at once has two rows, one for each.
- A parent who opted out (recorded in `consents`) has no row and gets the Control value (warm).

Parent	Experiment	Group given	Message style they get
P-07	style-test-1	Test (Option B)	playful
P-08	style-test-1	Control (Option A)	warm
P-09	*none: opted out, no row*	*gets Control*	*warm*

5 The app stamps every event

`App, every time · events · What happened`

One row per event: each nudge sent, habit logged, "Not today" tap, Wrap Up and so on. Each row records the settings version and the experiment group that parent was in at that moment.

- It **keeps growing**: before, during and after experiments, because events are recorded from day one. Rows are only ever added, never changed.
- Rough size: about 10 events per parent per day. That's about 120 rows a day in week one, and about 3.6 million a year at 1,000 parents.
- After an experiment stops, new events are simply stamped without it.

Parent	What happened	When	Settings version	Experiment group
P-07	Nudge sent	Wed, week 4, 8:02pm	version 1	style-test-1: Test (playful)
P-07	Habit logged	Wed, week 4, 9:15pm	version 1	style-test-1: Test (playful)
P-08	Nudge sent	Wed, week 4, 7:01pm	version 1	style-test-1: Control (warm)

6 Telemetry compares Control and Test

`After the stop · reads events`

Because every event carries its stamp, Telemetry can split any metric by group. For example, metric 8 (logs after a nudge) for parents stamped **Test (playful)**, compared with parents stamped **Control (warm)**. If playful wins, you can make it the new Control value for everyone: a change for everyone (way A in 2.5), which saves settings version 2.

2.5 Two ways to change a value

A. Change it for everyone

Example: move the Wrap Up default (row 3) to 7:30pm. Possible from week one.

1. YOU
   Tell me the change and why.
2. CLAUDE
   Edits `src/settings.ts` (and the plain-English list), runs the tests, opens a pull request.
3. YOU
   Review and merge.
4. APP
   When the new code goes live, it adds **settings version 2** with your note. From then on, every event is stamped "2".

B. Try it on some parents

Example: playful vs warm messages (row 11). From alpha only (rule 11.3).

1. YOU + CLAUDE
   If the option isn't in the code yet, add it by pull request. For a style, the playful wording is written and reviewed then.
2. YOU
   Add the experiment, its options, and a **start** row. (How you do this safely is designed before alpha; never from a web page that holds the master key, rule 11.7.)
3. APP
   Puts each parent who hasn't opted out in Control or Test at random, and adds an `assignments` row. Their group never changes.
4. APP
   Stamps every event "style-test-1: Test (playful)" or "style-test-1: Control (warm)".
5. YOU
   To finish, add a **stop** row. Everyone goes back to the Control value, and Telemetry compares Control and Test.

What way A's step 4 looks like: a new settings version

Example: in week 5 you move the Wrap Up default (row 3) from 8:00pm to 7:30pm, **for everyone**.
This is not an experiment. The app saves a new full copy of all 29 values. Only one is different.

Version 1

In force from trial start · "Starting values"

29 values

1	Default slot times	
2	Behaviours a parent may pick	
3	Weekly Wrap Up default time	8:00pm
4	Slots per schedule	
5	Slot length and range	
6	Times a week for deliberate behaviours	
7	Examples in a message	
8	Frequencies offered	
9	Hidden ideal and floor	
10	Reward numbers	
11	Message style	
12	Framing	
13	Message length and emoji	
14	Button wording	
15	Thank-you style	
16	Days deliberate behaviours are nudged	
17	Rest after a streak	
18	Defaults pre-selected or not	
19	Priority order	
20	Timing within a slot	
21	Use of the parent's own context	
22	Photo or description prompt	
23	Claude model used	
24	Badges and streak rules	
25	Nudge messages per day	
26	Gap between slots	
27	Note length limit	
28	Time zone	
29	At least 1 slot	

Version 2

In force from week 5 · "Wrap Up default moved"

29 values

1	Default slot times	
2	Behaviours a parent may pick	
3	Weekly Wrap Up default time	7:30pm
4	Slots per schedule	
5	Slot length and range	
6	Times a week for deliberate behaviours	
7	Examples in a message	
8	Frequencies offered	
9	Hidden ideal and floor	
10	Reward numbers	
11	Message style	
12	Framing	
13	Message length and emoji	
14	Button wording	
15	Thank-you style	
16	Days deliberate behaviours are nudged	
17	Rest after a streak	
18	Defaults pre-selected or not	
19	Priority order	
20	Timing within a slot	
21	Use of the parent's own context	
22	Photo or description prompt	
23	Claude model used	
24	Badges and streak rules	
25	Nudge messages per day	
26	Gap between slots	
27	Note length limit	
28	Time zone	
29	At least 1 slot	

Only row 3 changed. Rows 1 to 2 and 4 to 29 are identical copies.

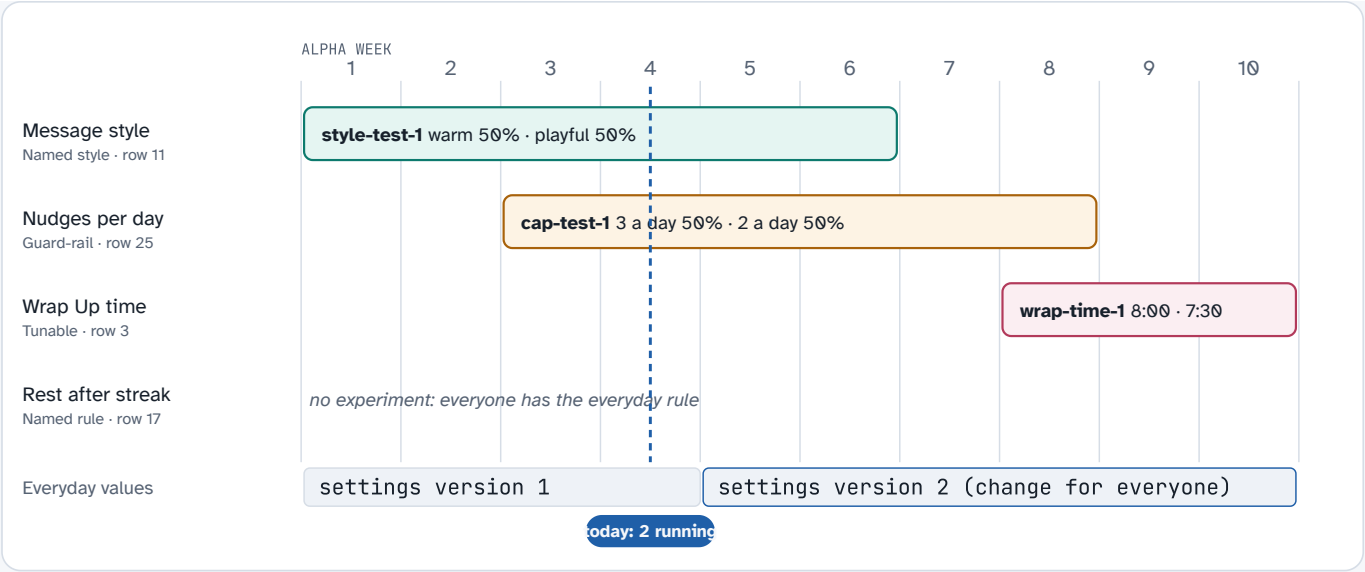
1 value changed, 28 unchanged. Version 1 is never edited. From week 5 every event is stamped "version 2", so later you can tell which Wrap Up default a parent had when something happened. If an experiment were running at the time, it would simply carry on, on top of version 2.

PART 3 · YOUR QUESTION 2

How to visualise if 2 or more experiments are running at the same time?

Read it like a train timetable. Each **row is one value** from the table in 2.2, and each **bar is an experiment** on that value.

- Bars in the **same row never overlap**: only one experiment per value at a time (rule 11.7).
- Bars in **different rows can overlap**, which means two experiments are running at once.
- The bottom strip shows the everyday settings version.
- The whole picture is drawn from `experiment_changes`, `experiment_options` and `settings_versions`.



Read straight down the dashed "today" line (week 4): **message style** and **nudges per day** are both running, and settings version 1 is in force. Each parent is in **one option of each running experiment**, so there are four combinations. Because the splits are random and separate, each combination gets about a quarter of the parents.

Parent P-07 in week 4 — their event stamp

```
settings_version: 1
variants:
  style-test-1 = playful
  cap-test-1   = 2 a day
```

Parent P-08 in week 4 — their event stamp

```
settings_version: 1
variants:
  style-test-1 = warm
  cap-test-1   = 3 a day
```

> **Running experiments together.** Two at once is fine when the values are unrelated. If they could affect each other (a playful style might matter more when there are fewer messages), Telemetry should check all four combinations, not each experiment alone. With small groups, run **one or two at a time**.

PART 4

Decisions: made, still open, parked

Made on 1 Nov

TOPIC	DECISION
Side-chat review	Applied, with Claude's three changes: "counts for which week" stays in `week.ts` (one copy of rule 7.6); frequency stored as a plain name checked in code; three narrow edits allowed, each writing an event.
Photo that fails the check	Stored and shown to the parent in the Web app; the parent is asked to self-report, and the log is recorded as self-reported.
Missing tables	Added now: `invites`, `holiday_dates`, `connection_answers` (plus `event_types` from the review).
Child's age	Birth year per child, optional, in `children`. Created now, empty in week one, asked from alpha. Ages 0 to 25 accepted; impossible years refused at sign-up and in code; ages 0 to 6 and 13 to 25 flagged in Telemetry.

TOPIC	DECISION
Opt-out (was E)	Recorded in `consents`, not `assignments`.
Build order (was G)	14 tables now; the four experiment tables before alpha; album entries and badges with Variable rewards.
New settings versions (was B)	Automatic, with the fingerprint protecting against duplicates.
Picks history (was C)	Kept, with a "stopped at" date.
Data about the parent (was D)	As listed, plus the optional birth year per child. Parents are warned that descriptions are free text.

Still open

"Agree" is enough where I've made a suggestion.

A **Where do the "may try" options live?**

In code, changed by pull request, or typed into the experiment tables without one.

Suggest in code. It keeps rules 1.4 and 11.10, and every option parents see was tested first.

F **The "what's running" timeline**

Add it to the backlog as an owner-only page before alpha?

Suggest yes (backlog, not now).

H **One set of kinds instead of two**

Rule 1.4 labels values guardrail, tunable or trial-plan; rule 11.10 uses number, named style or named rule. The five kinds in 2.1 merge them.

Replace the labels in rules 1.4 and 11.10 with the five kinds (dated rule change), or keep them in this draft only?

I **Trial-plan values**

Values such as the 5-week trial length aren't in the 29.

Leave them out on purpose (they shape the trial, not a parent's experience), or add them?

J **Values in the code but not in the table**

Wrap Up on by default; the Wrap Up day (Sunday); fewest examples per nudge (3).

Add any of them?

K **Spelling**

"Guard-rail" (your table) or "guardrail" (the specs)?

Which should I use everywhere?

L

Two numbers in the wrong file

The frequency choices (row 8) and the 2-day floor (row 9).

Suggest moving them into `src/settings.ts` in the Datastore pull request (same behaviour, tests unchanged).

M

Second tap on the weekly connection question

If a parent answers twice in one week, which answer counts?

Suggest the first one counts and a second tap is ignored, so the weekly number stays simple.

Parked until you design the Web app

- **How the Web app reaches the database:** through the server with the master key, or with limited per-parent access.
- **Parents' descriptions and Claude:** whether descriptions are sent to Claude to write messages, and what sign-up tells parents about it.

Draft 3 by Claude, 1 Nov 2026 SGT, for the owner of the Family Habit App. Includes refinement packs 1 and 2, the side-chat review, and the decisions on photos, missing tables and the child's birth year. Nothing is recorded in the specs or built yet; that starts when you approve this draft.